Since each hour has 60 minutes therefore contains 12 5-minute intervals, starting from minute 0 until minute 55. Therefore, a set of interval is $H_k = \{0, 5, 10, \dots, 55\}$ where the interval value is v_i for $i \in H_k$. Let define the hourly dispatch price as

$$x_h = \frac{1}{12} \sum_{i \in H_k} v_i \tag{1}$$

where h is the index of the hour, not necessarily be in the 24 hour limit.

However, I have a population of hours in a whole year. I want to estimate that population mean, hence

$$\theta = \frac{1}{N} \sum_{h=1}^{N} x_h \tag{2}$$

I will use stratified sampling, each stratum is block of hours in a day such as morning peak hours, working hours, evening peak hours, night hours, sleep hours to capture the characteristics of the data set. Therefore, there are 5 strata ($M = 5$). I want to create a sample of arbitrary size n. So my population mean now is,

$$\mu = \frac{1}{N} \sum_{d=1}^{M} \sum_{h=1}^{N_d} x_{dh} = \sum_{d=1}^{M} w_d \mu_d \tag{3}$$

where $w_d = N_d/N$, $\mu_d = \frac{1}{N_d} \sum_{h=1}^{N_d} x_{dh}$. The latter summation is the weighted average of strata means.

Let n_d be the sampling size of each stratum, where $1 \le d \le M$ and $n_1 + n_2 + \ldots + n_5 = n$